FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

TCG 6223- COMPUTER GRAPHICS

Trimester March 2025 (2510)

Project for **Team Ninja**

We hereby declare that all group members' names are correctly included. We certify that not part of this assignment has been copied from any other student's work or from any other source.

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Introduction

This project represents a simple little animated game built using C++ and OpenGL in the Code:: Blocks environment its called (Slime Vs Ghost). The game features was two main characters: a ghost and a slime, each with their own unique skills style and interactions within the environment.

The ghost glides fluently through the air with a fading effect, making it an eerie, floating presence. On the other hand , the slime bounces playfully across the ground just like a ribbit, adding a fun and lively contrast. Together, they bring the scene to life with basic but effective animation techniques.

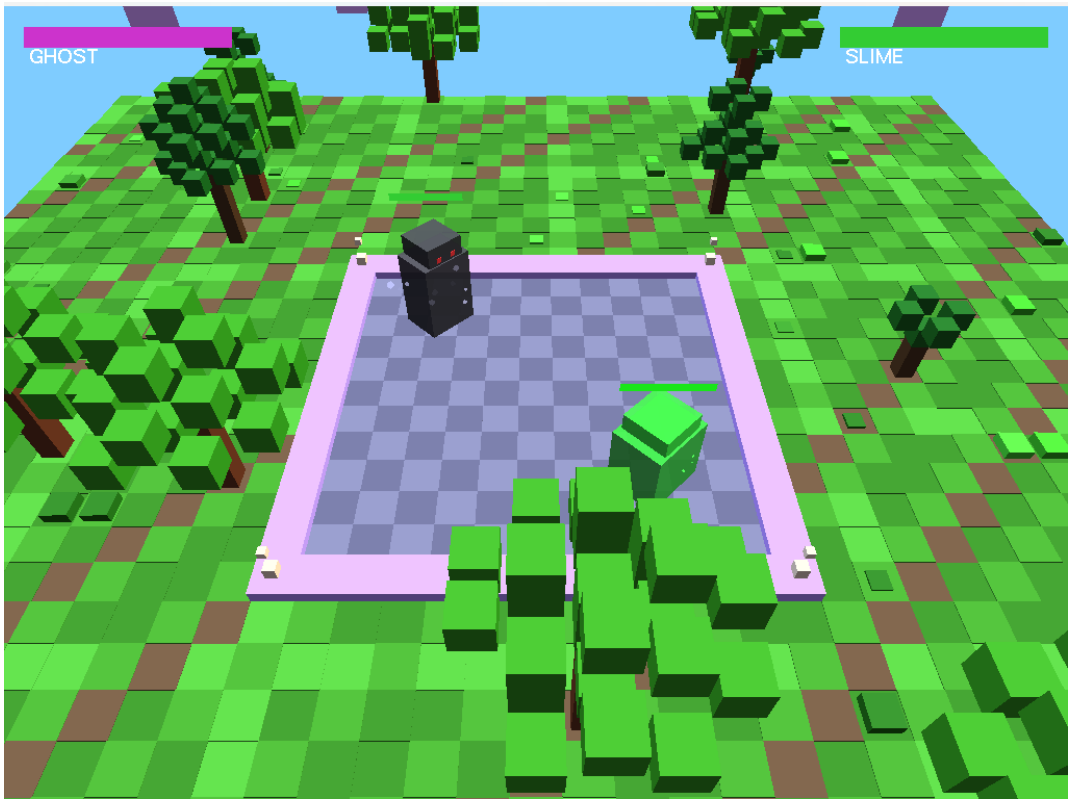


Figure.1

The goal of this project was to apply object-oriented programming principles to create modular and flexible , adjustable classes for each animated object. By separating basic physics and logic into different files (like Ghost.cpp and Slime.cpp), the project remains clean, organized, and easy to expand or modify.

Documentation of Virtual Model

3.1 Ghost Character

1. Description of Objects
The Ghost is a blocky, partially see-through character that utilizes a voxel style (e.g., Minecraft). It has the following parts:

- **Head:** Cube with glowing red eyeballs.
- **Torso:** Rectangular body can be made from stacked small cubes.
- **Arms & Hands:** Both arms and their hands are modifiable for animation.
- **Weapon (optional):** Cylinder or cube.
- **Particles:** Glowing and trailing particles when the ghost dashes, attacks, or becomes invisible.

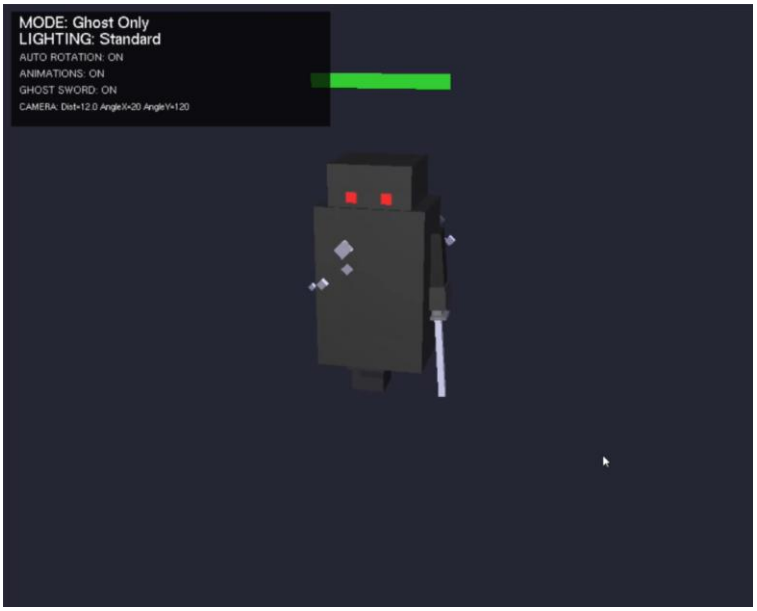


Figure1: Ghost

Special functions:

- can become invisible.
- has a sword attack.
- health bar is above the ghosts head.

2. Creation and Animation

Using:

- **glutSolidCube()** for most of the objects.
- **glTranslatef(), glRotatef()** to set the position and animation.
- Custom functions i.e. **drawVoxelCube()** and **drawGlowingSphere().**

Details of the animations:

- **Idle:** gentle motion up-down using sine waves.
- **Sword Attack:** arm and sword are rotating forward.
- **Dash:** moving very quickly with trailing effect using particle.
- **Invisibility:** uses alpha blending to fade in and out.

1. Ghost Character Attack Animation

This figure depicts the ghost character in an attack animation. The purple blades and glowing particles around the body indicate that some action is taking place in the battle arena.

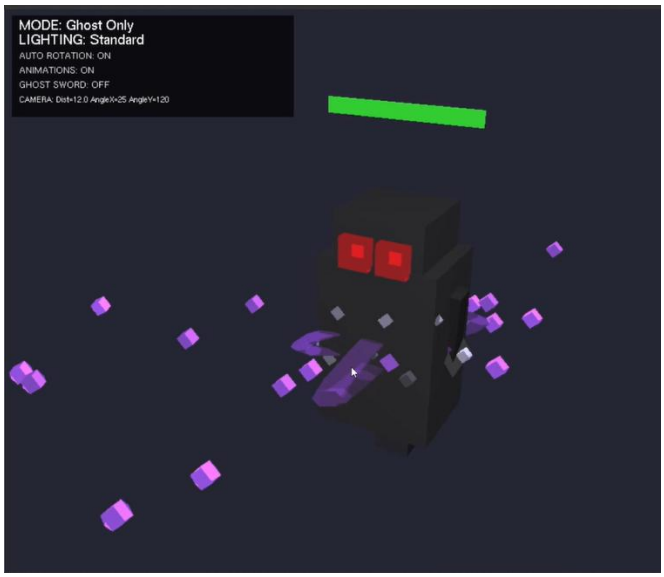


Figure2: Attack

2. Ghost Character Invisible.

This figure depicts the ghost character invisible. The body is semi-transparent and has a tint of edges and other less distinct colors while the other two most prominent components, the eyes and sword just have some light color defining them.

Figure3: Invisible

3. Drawing Boundaries (As Wireframe)
This figure depicts the ghost character as wireframe. The bounding cubes and skeleton nodes are visible indicating it is the design for animation and movement.

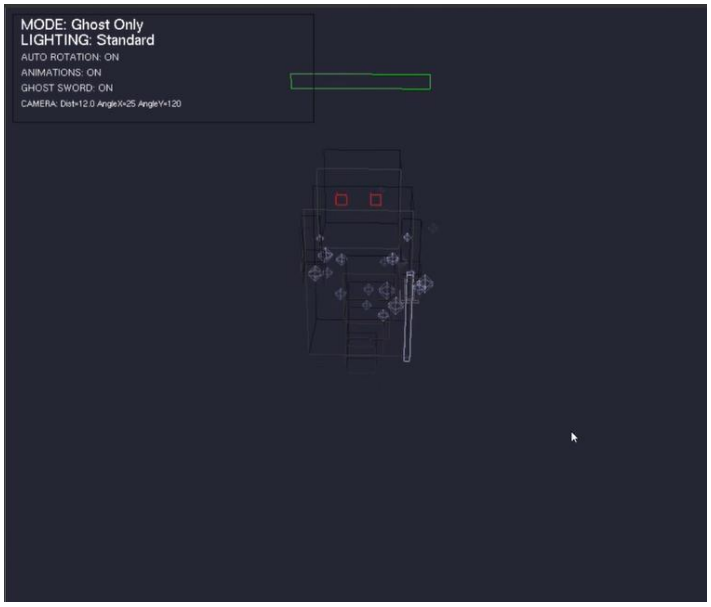
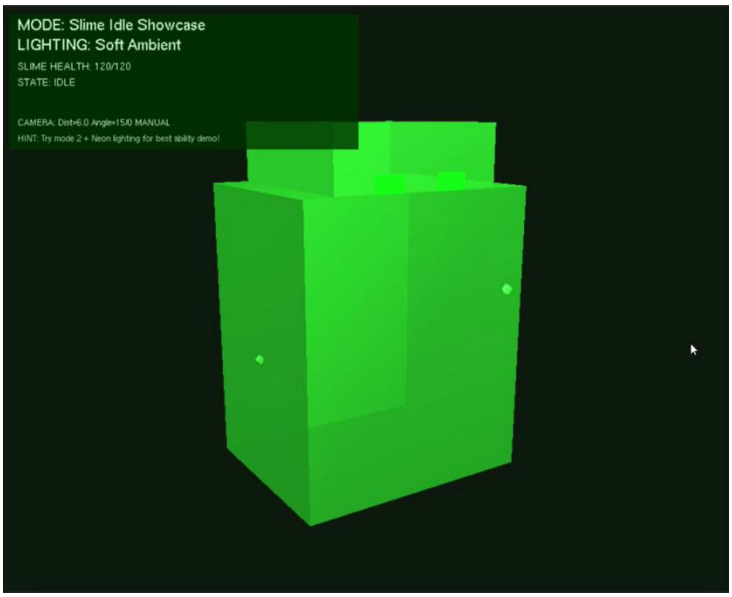


Figure4: Drawing Boundaries

3.2 Slime Character



1. Description of the Object

The Slime is a blocky, glowing green character using voxel-style graphics (think Minecraft). What is the Slime made of:

- Body - a collection of cubes that make a cylinder.
- Eyes - two small glowing dots or cubes on top of the body.
- Particles - glowing particles that surround the body, to create jelly-like visual effects.
- Health Bar - displaying health status above the head, indicating current health.

Special characteristics for the Slime:

- Can change its appearance and perform various actions ; bomb attack, shield, bouncing.
- There is a translucent look that appears while some of the abilities are active.

2. Build and Animation

Constructed of:

• many glutSolidCube() for most of the features along with using glTranslatef(), glRotatef() to place and animate.

• some custom functions, drawVoxelCube(), drawGlowingSphere().

Animation features:

• Idle - slowly floating around using sine waves.

• Bomb Attack - have the body dynamically scaling.

• Shield - to draw a translucent dome around the slime.

• Bounce - travelling through space up and down using oscillating movements.

Figure 1: Slime Wireframe

Here is the wireframe for the Slime's body only, represented as stacked cubes on top of each other. The wireframe illustrates how characters are made using modular/block parts.

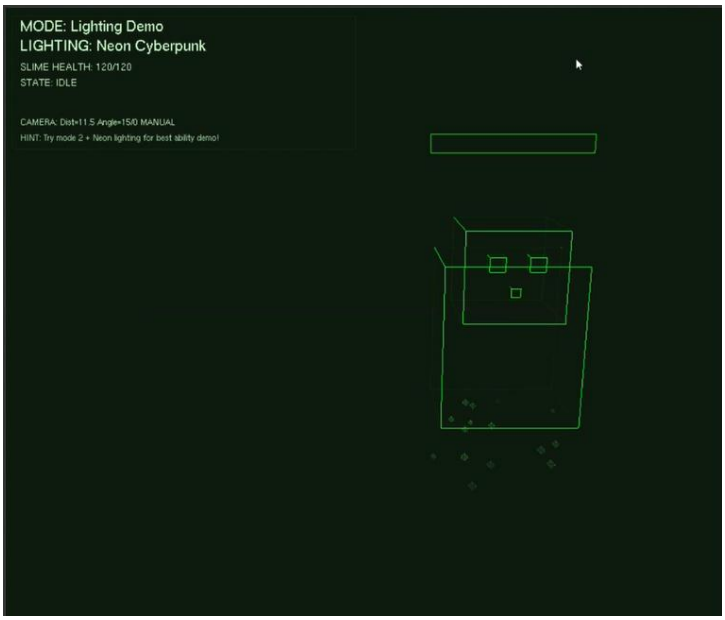


Figure1: Slime

Figure 2: Slime Attack

The Slime is attacking in this image. The Slime is shown firing glowing projectile balls (or orbs) towards the enemy, while a burst of purple particles displays the direction of the projectile's flight and impact with the enemy.

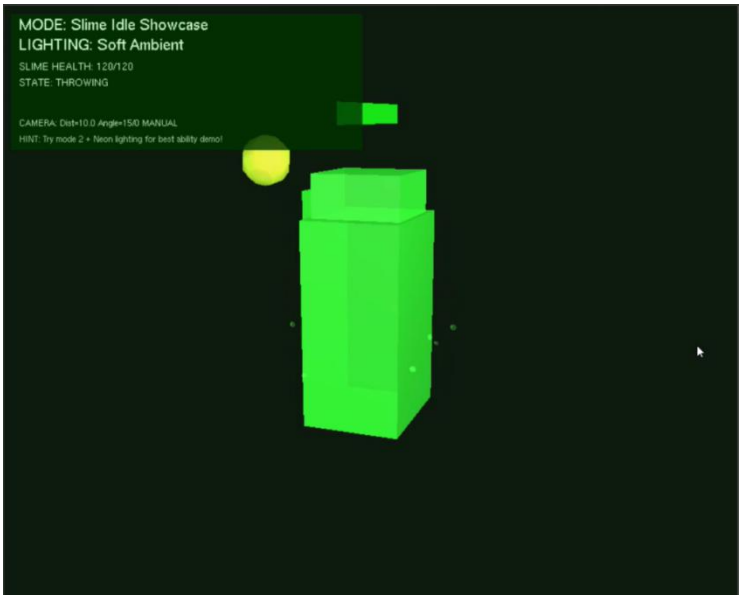


Figure2: Slime Attack

Figure 3: Slime Shield Activated

This image shows the Slime with its shield protection activated. The image depicts the shield effect as a partly-transparent shielding effect and elemental particles floating off the body, which could be described as a magical attack and defense.

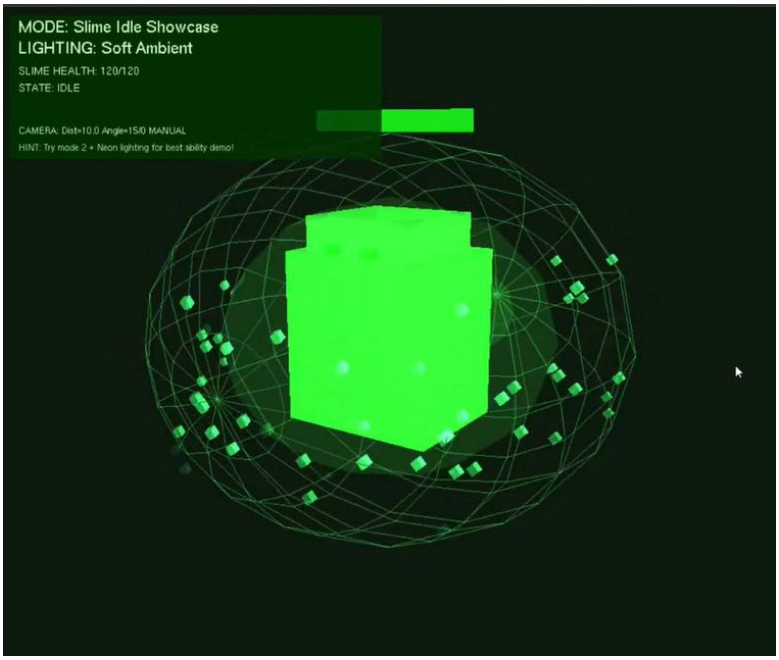


Figure3: Slime Shield

3.3 Environment - Trees & Nature

1. Building Objects Description
The Environment part of Trees & Nature has a goal to build a scene that is somewhat realistic and voxel-based when compared to other voxel games like Minecraft. The building components that are a part of this component are illustrated below:

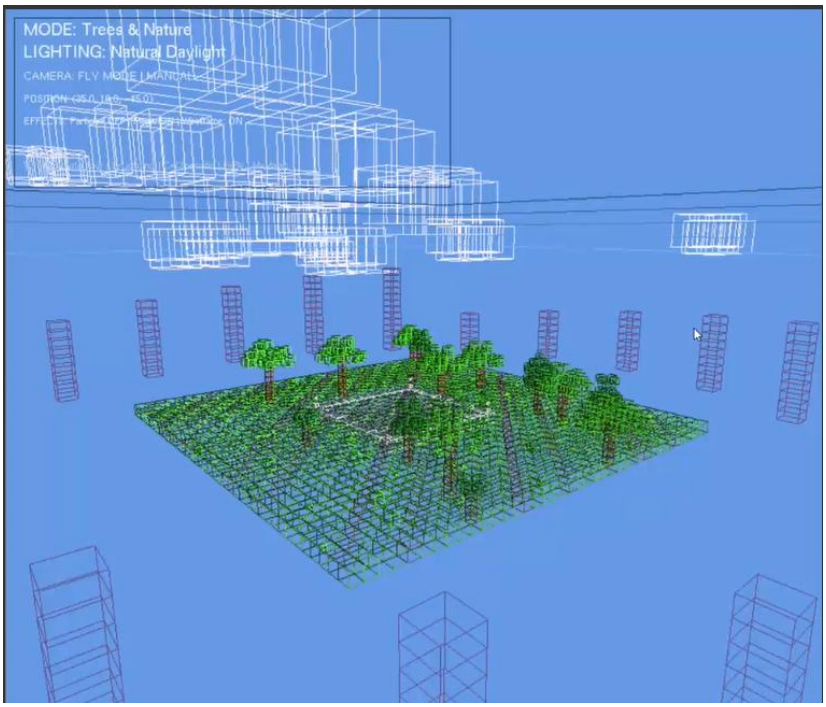


Figure1: Environment

- **Terrain** - A grid-based terrain that is built up in cubes.
- **Trees** - Cylinders for trunks and sphere or cubes for leaves/foliage.
- **Battle Arena** - A flat/open area where a character is in battle with glow or horizon marker.
- **Lighting Effects** - More complex lighting/modes to project a time of day or graphic representation of mood.

Special Features:

Many lighting modes (i.e. Natural Daylight, Mystical Twilight, Cool Moonlight)

Camera controls for Qi to orbit, zoom in / out, and rotate around the Scene.

Very configurable and scalable modular components.

Animation Specifications:

- Ground: Static, but we could do something dynamic for things like particle trails or deformation of the ground.
- Trees: Added to the ground at random locations using procedural generation.
- Battle Arena: Made with glowing outlines, used a lighting system to dynamically respond and visually demonstrate the area of battle.
- Light Modes: dynamically change the color and intensity of the selected light modes.

1. Cool Moonlight Lighting,

This figure illustrates the environment in cool moonlight lighting. The terrain is dark, but blue tones still dominate the visual setting, while terrain trees can still be located. The battle arena has engulfed the ambient blue light of the terrain's moonlight.

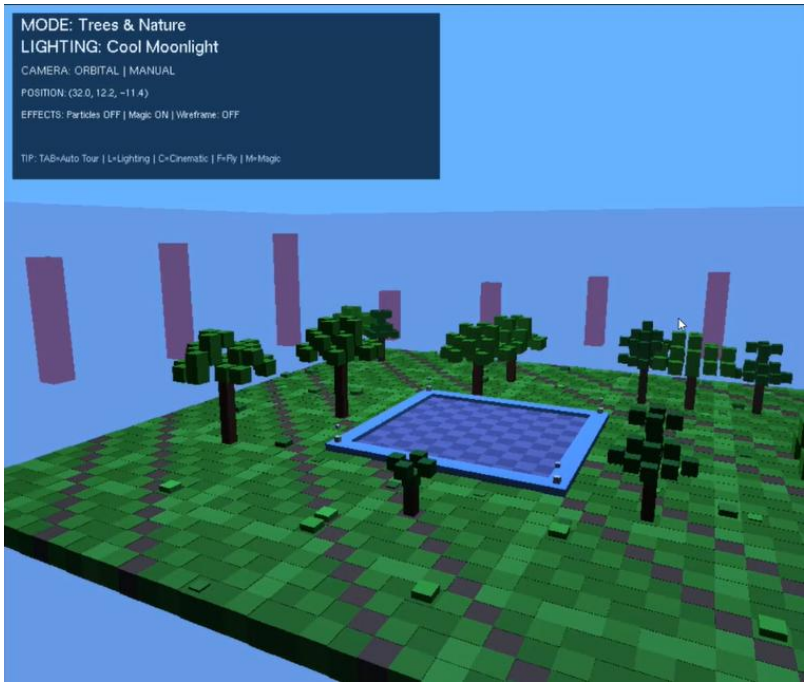


Figure2: Cool Moonlight

2. Magical Fantasy Lighting,

This figure illustrates the environment in magical fantasy lighting. Bright pink and bright purple tones dominate the scene; radiant trees and a sparkling battle arena provide the counterbalance to the magic of this lighting mode.

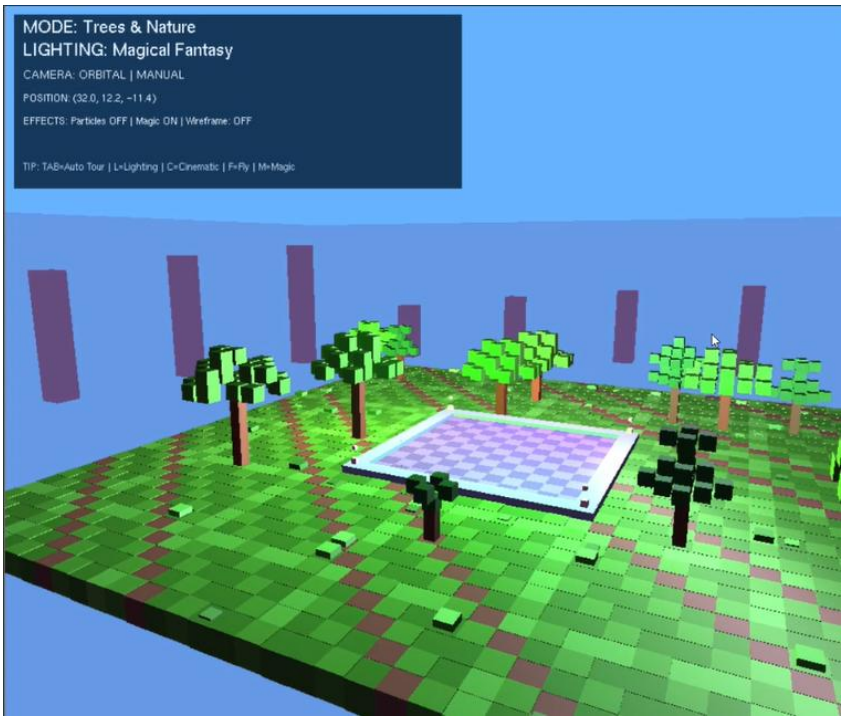


Figure3: Fantasy Lighting

3. Volcanic Glow Lighting

This figure shows the environment under volcanic glow lighting. Warm red-orange tones evoke lava, while the terrain appears like heat distortion off the lava, the trees are glowing, changing, and the battleground arena is molten.

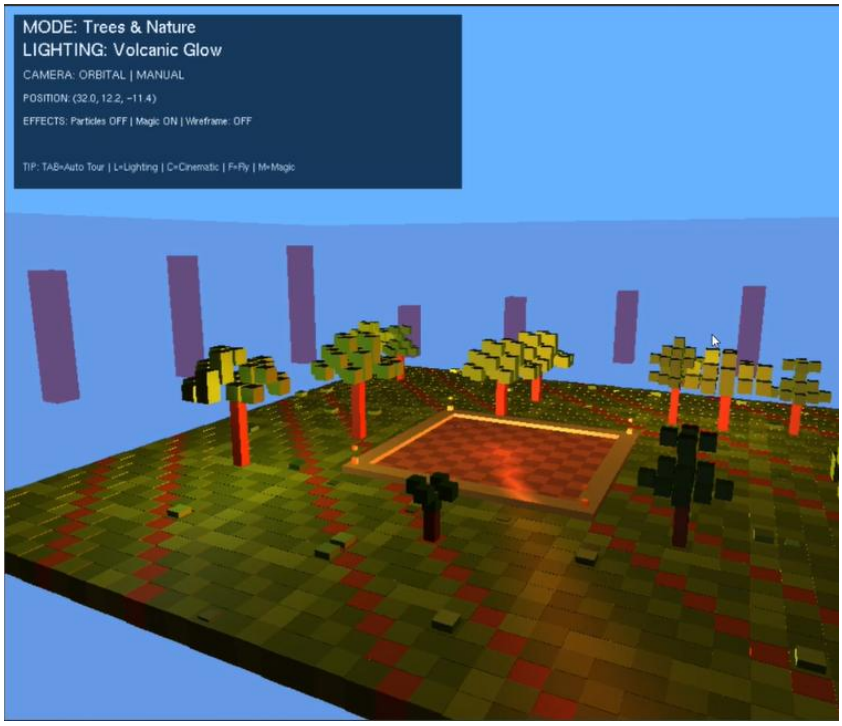


Figure4: Volcanic Lighting

4. Natural Daylight Wireframe

This figure shows the wireframe of the environment. The wireframe consists of axial outlines of cube-based terrain (cube map), cylindrical tree trunks, and a planar battlefield arena with boundaries.

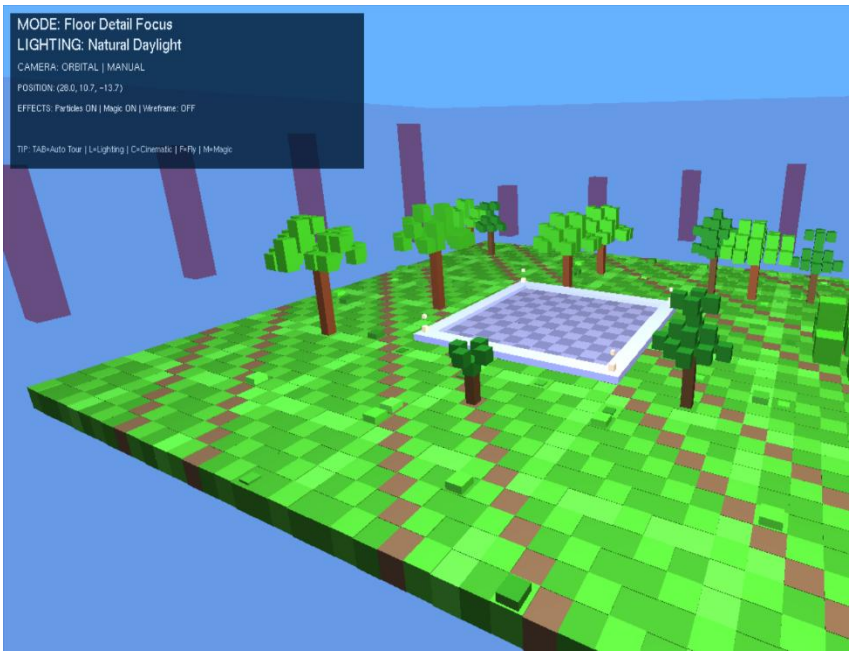


Figure5: Natural DayLight

User manual

Slime keyword

Botton	Function
↑	Move Forward
↓	Move backward
→	Move Right
←	Move lift
6	Slime boom
7	Slime Shield
8	Slime bounce dodge

Ghost keyword

Botton	Function
1	Dash attack
2	Shadow swipe attack
3	Ghost Shield
4	Ghost invisibility

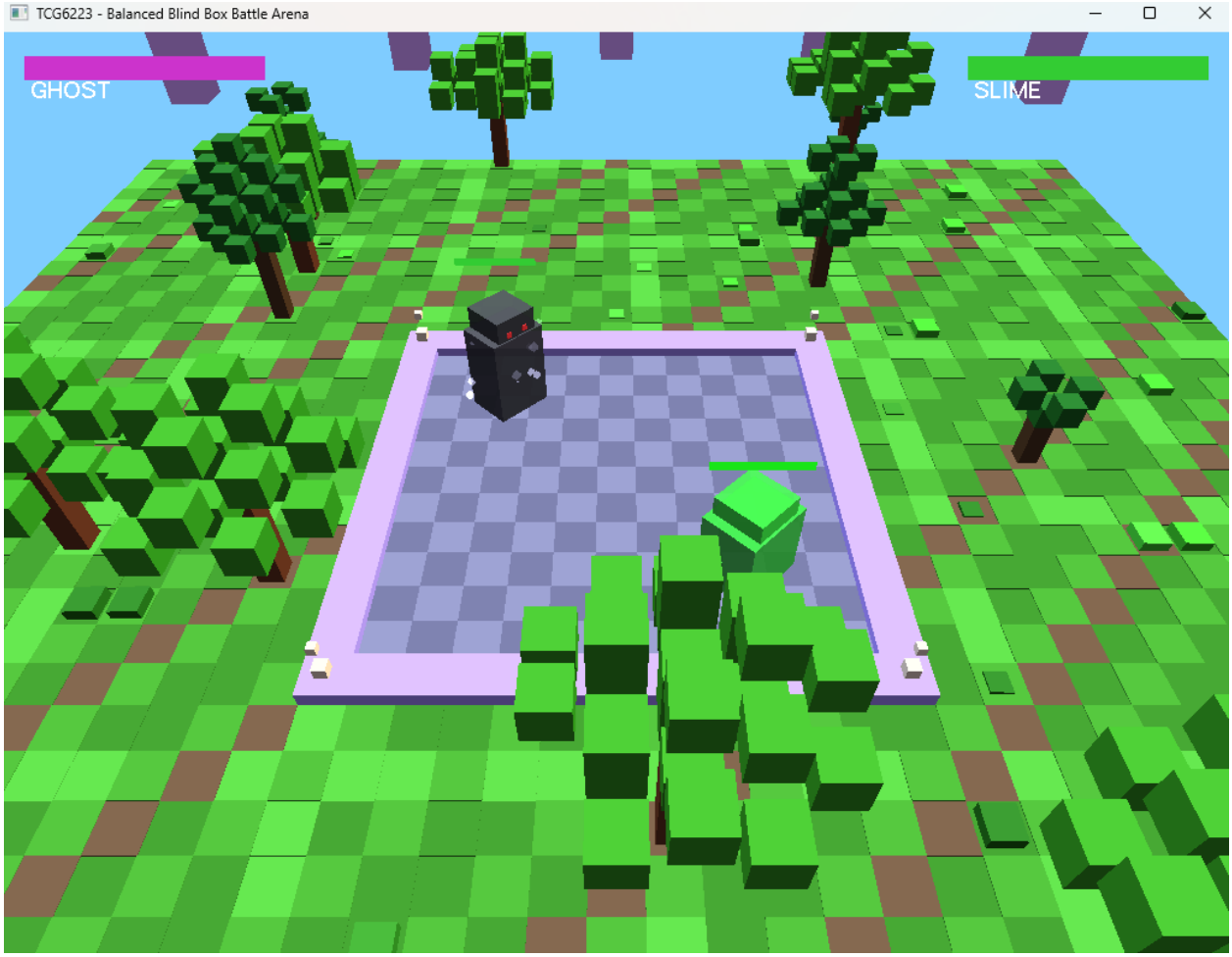
Battle System

Botton	Function
5	Reset battle
0	Pause/Resume
ESC	Exit

Camera control

Botton	Function
W	Move Forward
S	Move backward
D	Rotate Right
A	Rotate left
Q	Move camera down
E	Move camera up
R	Reset the camera position
L	Cycle lighting

Battle Arena Overview



Description:

A top-down view of the battle arena, illustrating the Ghost and Slime characters in opposition. The arena has a glowing border with light pillars on all four corners. The floor has a checkerboard pattern. The rest of the environment includes some trees, grass sprigs, and clouds in the background. **Key features:**

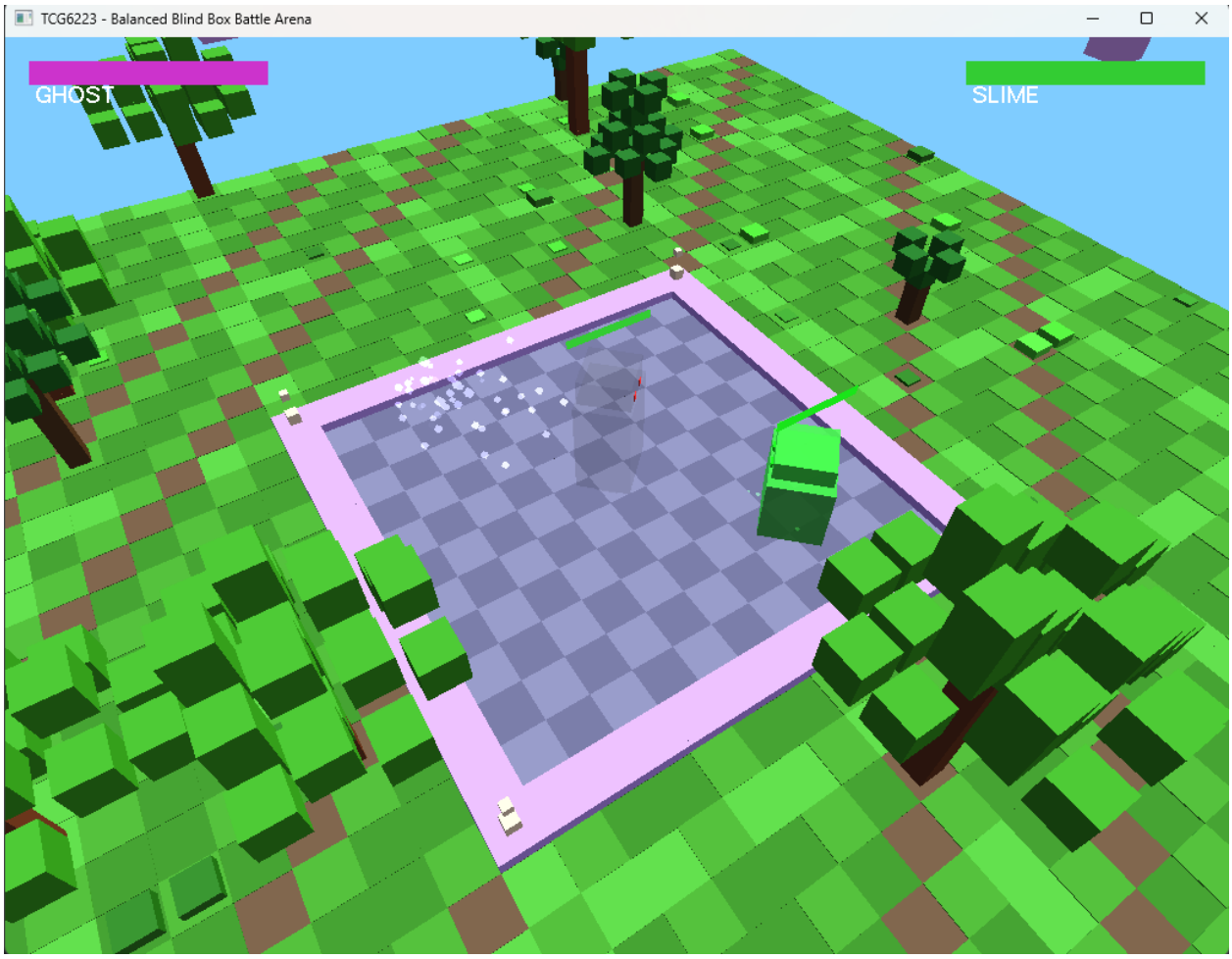
• Ghost (dark, cubic, glowing red eyes) is on the left.

• Slime (green, cubic, glowing green eyes) is on the right.

• Glowing border stones that pulse in and out.

• Skybox has mountains in the background**.**

Ghost Using Shadow Swipe Attack



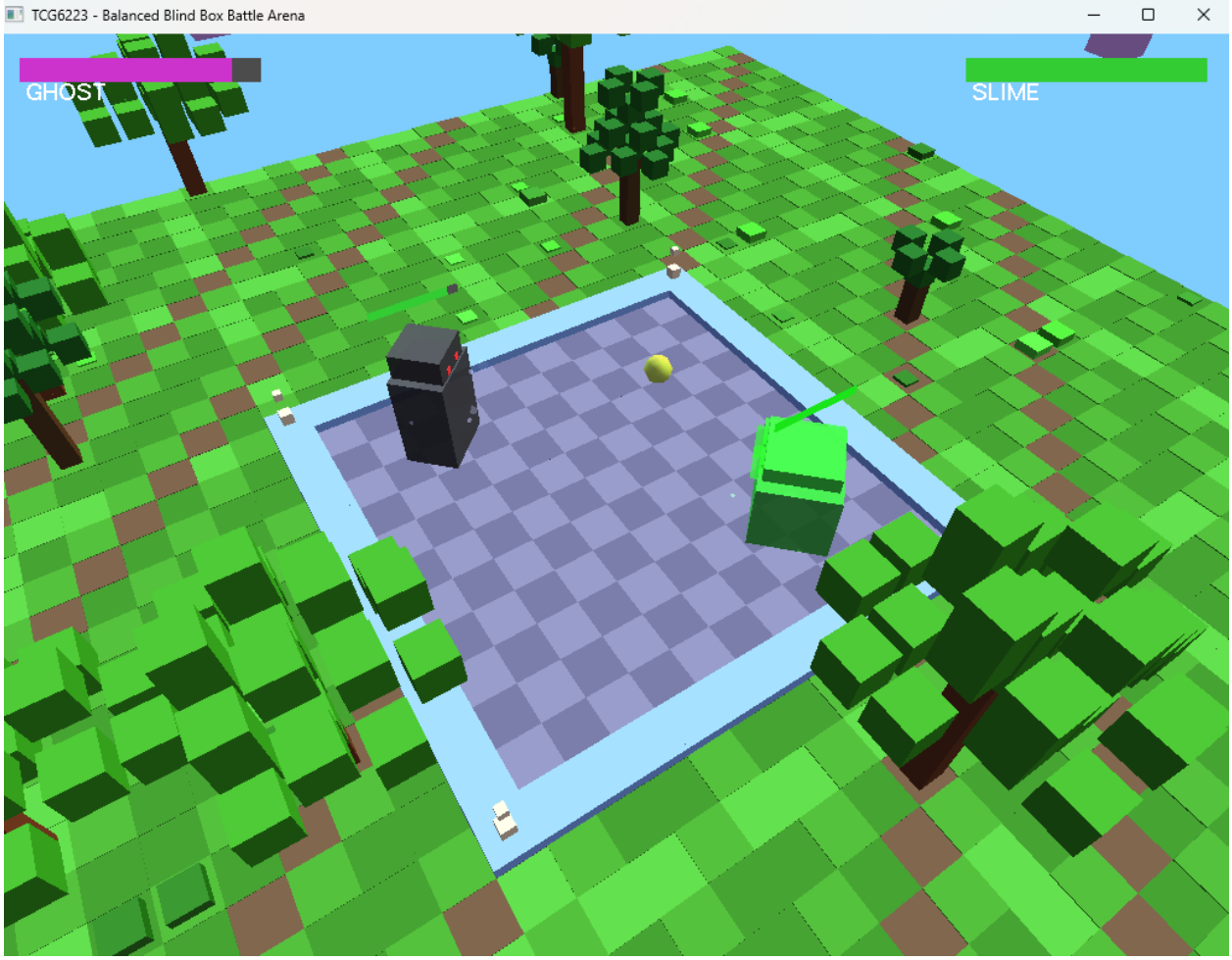
Description:

The Ghost is currently attacking, doing a "Shadow Swipe" where dark energy waves are extending from the attack. The Slime is pushed back slightly, and there is an explosion of the particles at the point of impact.

Key Features:

- The Ghost's eyes look very red.
- Dark purple energy rings.
- The Slime is slightly squished from impact.
- Particles from the magic effect near the Slime.

Slime Throwing a Bomb



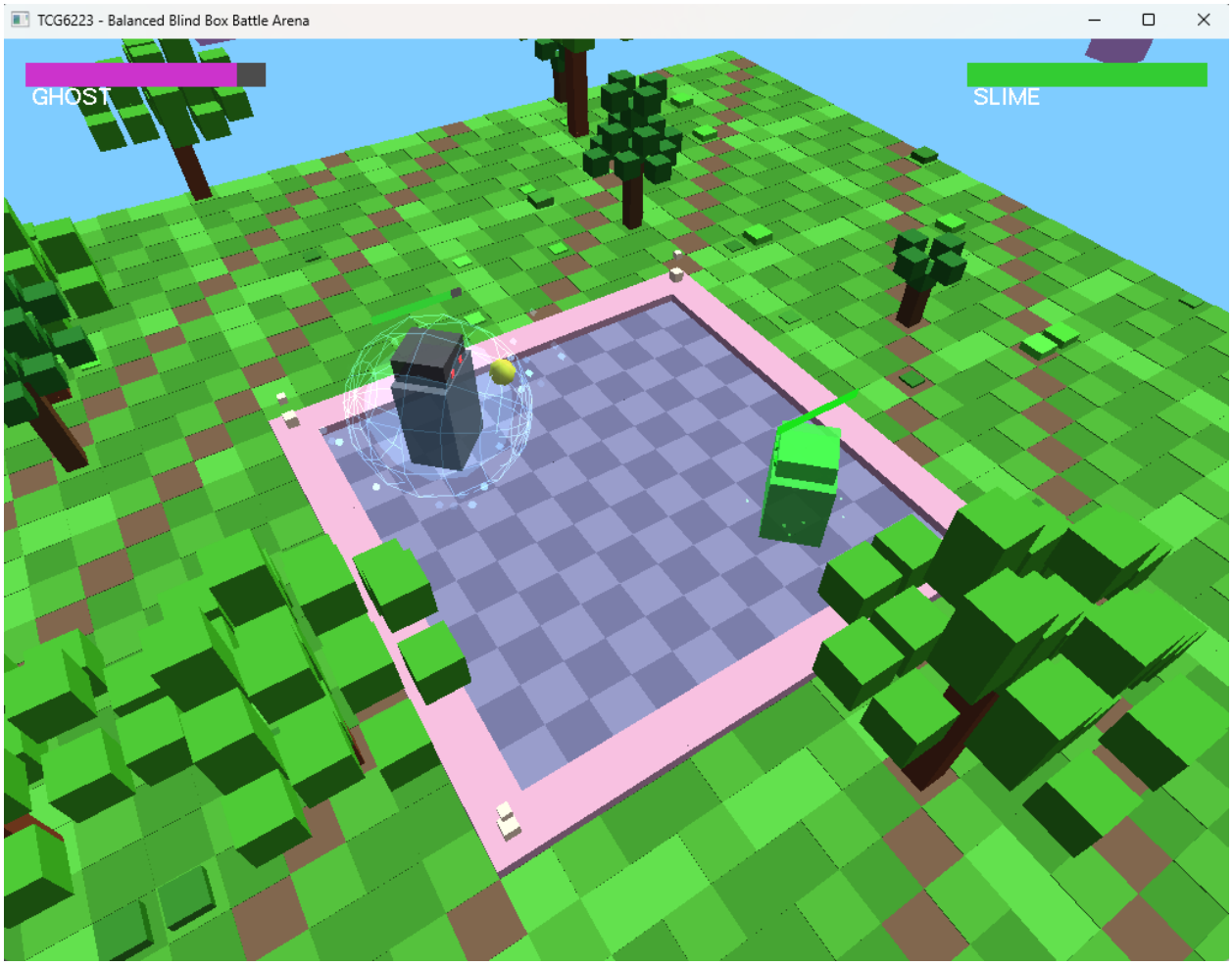
Description:

The Slime is currently launching a glowing green bomb to the Ghost. The bomb is sent through the air with a yellow core and vigorous spin. The Ghost has a blue shield activated so he can protected him self .

Key features:

- Slime squished (squish animation) while throwing.
- Glowing green bomb with an appropriate trajectory arc.
- Ghost blue shield pulsing.
- Slime trail particles on the ground.

Ghost Shield Activated



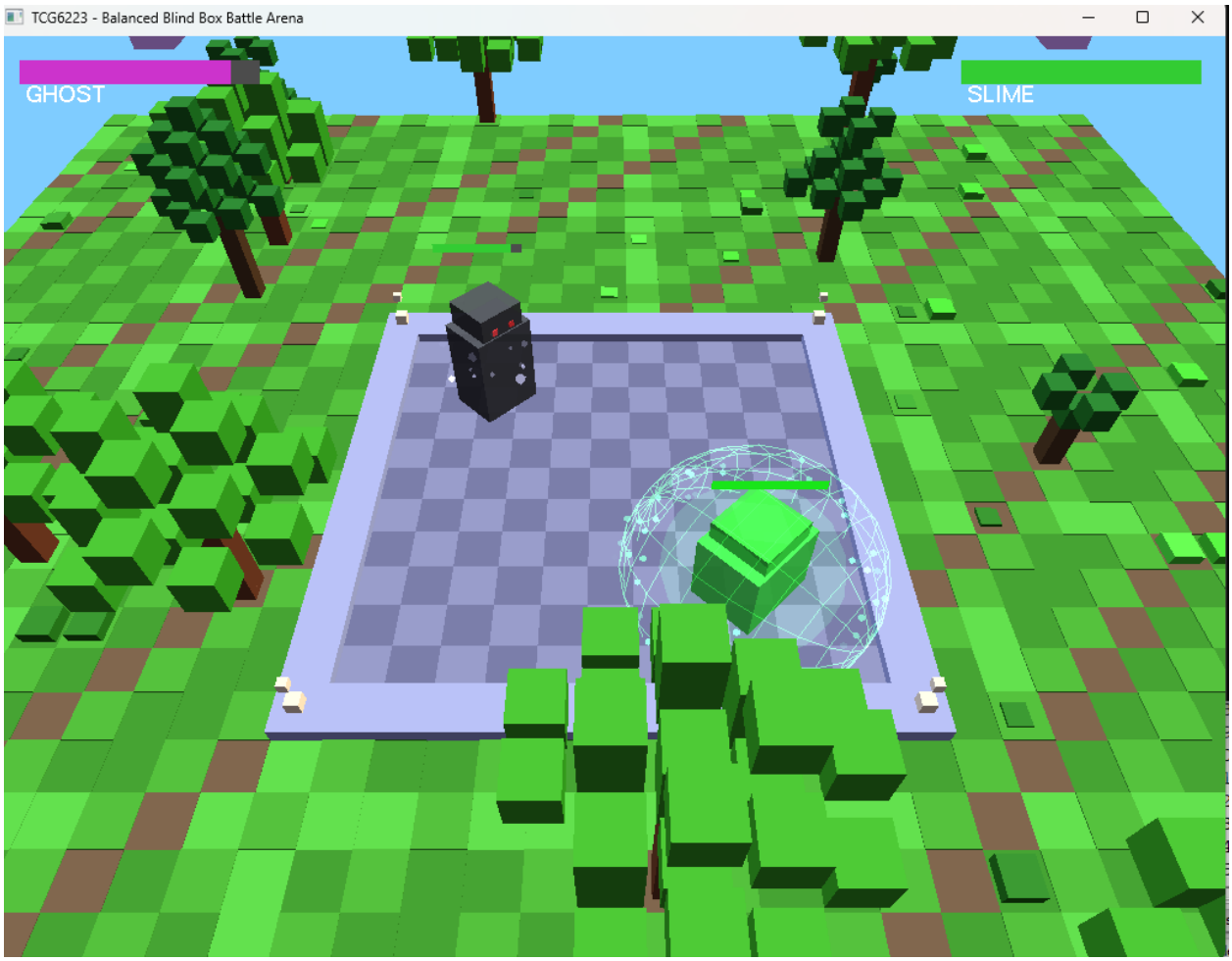
Description:

The Ghost has a bright blue hexagonal shield surrounding it, pulsing with energy. The green have throw his ball but there is no damage on Ghost.

Key Features:

- Blue shield with wireframe and inner glow.
- Ghost slightly enlarged while shielded.
- Slime has been activate his attack.
- Shield particles floating around Ghost.

Slime Shield Activated

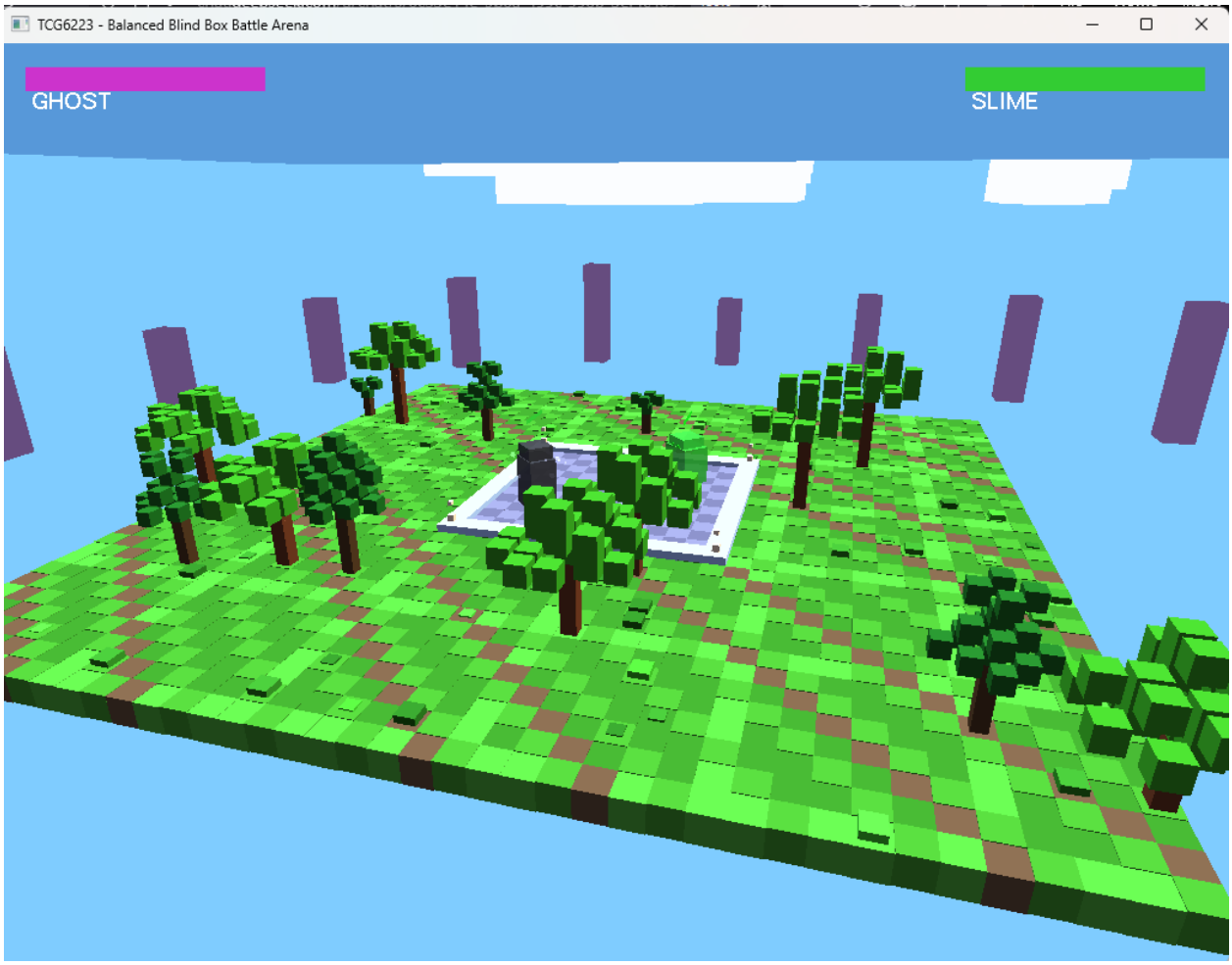


Description:
The Slime shows in a bright green hexagonal shield, pulsing with energy. The Ghost is nearby, revving up for an attack.

Key Features:

- Green shield featuring a wireframe appearance with an inner glow
- Slime scaled slightly bigger when shielded
- Ghost radey to use the shadow swipe to attack
- Shield particles are floating around the Slime

Environment Details (Trees & Skybox)



Description:

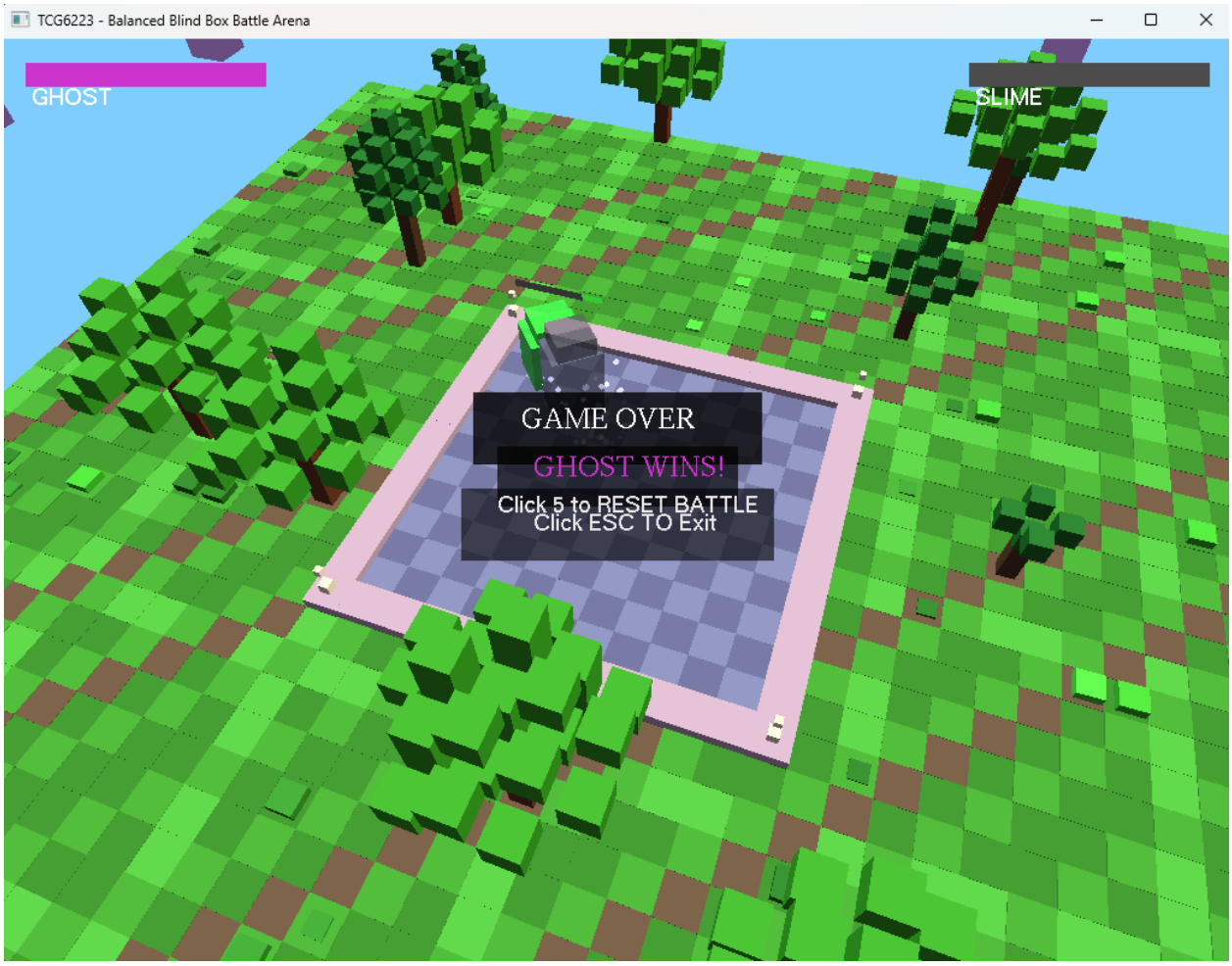
A side perspective of the environment that reveals the voxel-based trees, grass patches, and skybox with clouds. In the distance, the battle arena is visible.

Key Features:

- Oak and pine trees with cubic leaves.
- Animated grass fluctuation
- .Floating clouds that appear to be moving.

  Sky gradient from blue at the bottom to NASA approved dark blue at the top

Game Over Screen (Ghost Wins)



Description:

The Ghost is displayed in victory while the Slime is displayed in defeat (health bar is empty). The UI has "GAME OVER!" and "GHOST WINS!" in purple fonts.

Key Features:

- Ghost glowing with very high intensity.
- Slime flattened and darkened.
- Rectangular semi-transparent black UI panel with victory text.
- Arena border still pulsing in back.

YouTube Demonstration link

https://youtu.be/DFAask1rXgU